DPI Script User Guide

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DPI Script User Guide
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Preface

This guide describes how an Field Engineer or a Network Operator can enable DPI Script to add or update classification support for applications and protocols without any software update or a restart on MCC.

Audience

This guide is for Field Engineers and Network Operators who have a good understanding of network protocols and DPI.

The procedures in this guide require you to understand and follow the safety practices at your site, as well as those identified in this guide.

How to Use This Guide

This guide contains the following chapters and appendices:

Chapter/Appendix	Describes
	Introduces the DPI Script.
Configuring a DPI Script	Provides information required to configure a DPI Script before creating one.
Creating a DPI Script	Provides information to create a DPI Script.

Terms and Acronyms

For a comprehensive list of terms and acronyms used in Affirmed Networks documentation, please see *Affirmed Networks Terms and Acronyms*.

Conventions

This guide uses the following conventions, when applicable:

Description	Convention	Examples
Command names, keywords, dialog boxes, buttons, icons, menu	**Times New Roman bold font**	Use **configure_ha** to configure... Enter **y**. To continue, press **Enter**.
Selecting a menu item	**Menu => Option**	Select **Security => Group Management**
Variable parameters for which you supply values	***<courier bold italic blue font>***	***configure_dr --host <host name>***

<table>
<tr><td></td><td></td><td></td></tr>
<tr><td>User input</td><td><code>Courier bold blue font</code></td><td><code>cd /opt/Affirmed/NMS/bin</code></td></tr>
<tr><td>Screen messages, system output, sample source code</td><td><code>Courier regular blue font</code></td><td><code>Checking for License...</code></td></tr>
<tr><td>Required keywords and arguments in square brackets</td><td>[ ]</td><td><code>cluster [id] node [id] reboot</code></td></tr>
<tr><td>Optional keywords and arguments in curly brackets</td><td>{ }</td><td><code>{cdr-file-name}</code></td></tr>
<tr><td>Keywords separated by the pipe symbol. Requires you to specify only one item from the set.</td><td>|</td><td><code>Is this correct? [y|n]</code></td></tr>
<tr><td>Keywords representing a single element in angle brackets</td><td>< ></td><td><code><host name></code></td></tr>
<tr><td>Asterisk after a CLI object indicates a required object</td><td>*</td><td><code>name*</code></td></tr>
<tr><td>Variables, book and chapter titles, file path names, new terms, and emphasized text</td><td>Times New Roman italic font</td><td>See the Acuitas User's Guide.
Workflow is an application running on the network element.</td></tr>
</table>

Note: *Additional information that may apply to the subject text.*

Tip: Helpful suggestions.

Caution: Proceed carefully to avoid possible equipment damage or data loss.

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Chapter 1

Overview

The DPI script enables a Field Engineer or a Network Operator to add or update classification support for applications and protocols without any software update or a restart on MCC. The DPI script validates user-defined signatures and uses image management techniques to load the packages dynamically, which avoids application restart and the service performance is increased.

Allows Field Engineers or Network Operators to add the following using DPI script:

- Application
- Protocol
- Extending PACE2 App Classification through Custom Classification

  Example: When PACE2 detected an application, say “X-VPN” flows are not getting detected and a new pattern is identified, you can extend “X-VPN” application detection via custom classification.

- Extending PACE2 Protocol Classification through Custom Classification

  Example: PACE2 detected a protocol, say “WireGuard” flows are not getting detected and a new pattern is identified, you can extend WireGuard protocol detection via custom classification.

The DPI script supports adding a maximum of 200 signatures based on the below parameters:

- HTTP hostname
- HTTP User-Agent
- HTTPS Client Hello SNI
- HTTPS Certificate Common Name

Advantage

- Adds DPI Signature patterns using the script and loads it on to MCC without a restart.
- Reduces time taken to support DPI Signatures using the PACE2 Custom Defined Classification (CDC).
- Enables to release of a new DPI Signature without a new PACE2 library.

- Supports current features of Custom Classification framework when the script-based approach is followed.
- Adds new script tokens using Affirmed DPI signature with continuous feedback from the Field Engineers.
- Submits script to Affirmed DPI team to add support for new patterns identified.

Limitations

Following limitations exist in the DPI script:

- Supports only one attribute per application/protocol signature.
- Protocol attributes added in the signature are not considered.
- Pattern included in the signature is not compared against existing data.
- Pattern search is not limited to the meta-data type mentioned in the signature.

  Example: If the user gives “abcdef” as Http user-agent pattern and if the same pattern is present in Http hostname, then the user-agent-based script signature results in a match.
- Script signatures are not adjusted during dpisig/MCC upgrade or downgrade.

  Example: If an application classification is added to script and if the same application is later supported using new dpisig, then the script-based signature is not disabled/discarded as part of dpisig upgrade.
- Command to see the fastpath loading status of DPI Script Signatures is not included.
- No tool or script is included to extract the DPI Script compiler.
- No offline tool to validate script signature using pcaps before loading to MCC.
- DPI Script compiler does not extend PACE2 application/protocol when there is a space in the PACE2 Application/Protocol Name.
- The value set for "affirmed-signature-extension" is not considered while compiling using the DPI Script compiler.
- DPI Script compiler accepts more than 256 Http rules and SSL rules.
- DPI Script compiler associates rules belonging to different apps to a single app when a combination of append and upgrade is performed.
- DPI Script PDL rule written without an attribute fails to compile.

Chapter 2

Configuring a DPI Script

By default, the DPI script feature is disabled in the MCC. To enable, set the value of the below parameter in the configuration:

```
data-path dpi dpiscript script-classification <disabled |
enabled>
```

> ***Note:*** *In the disabled state, user-defined signatures are loaded on the system but the classification/detection phase does not consider these signatures.*

Configure below parameter to enable/disable HTTP meta-data-based classification. This parameter takes effect only when “data-path dpi dpiscript script-classification” is enabled:

```
data-path dpi dpiscript http-classification <disabled | enabled>
```

Configure below parameter to enable/disable HTTPS meta-data-based classification. This parameter takes effect only when “data-path dpi dpiscript script-classification” is enabled:

```
data-path dpi dpiscript https-classification <disabled |
enabled>
```

Chapter 3

Creating a DPI Script

Follow the below steps to create a DPI Script:

- Collecting pcaps
- Analyzing pcaps and Capturing Patterns
- Script Syntax and Parameter Description
- Compiling a Script
- Loading a Script on the MCC

Collecting pcaps

This section describes the steps required to collect pcaps using the required application on the UE.

To collect pcaps:

1. Uninstall existing internet traffic generating apps like YouTube, Facebook, Instagram, Games, VPNs from the UE.
2. Install the required application on both Android and iOS devices.
3. Clear the subscriber and disconnect UE from MCC (turn off Wi-Fi or disable mobile data).
4. Close all the open apps on the UE.
5. Start the packet capture on the GI.
6. Create subscriber and attach UE.
7. Open only the installed app on the UE and perform a subset of options per iteration like Login, browse the content, Logout.
8. Clear the subscriber and stop the packet capture.
9. Copy the pcaps generated on GI side and rename them to reflect the actions performed on the app like <YYYY-MM-DD>_<AppName>-<version>_<UE OS>-<OS version>_allow_ <actions>.pcap (ex: 2019-09-02_maxdome-4.1.0_Android-8.1.0_allow_ Login+BrowsingTheList+Logout_1.pcap)

Analyzing pcaps and Capturing Patterns

This section describes the steps required to analyze pcaps using pcap analysis tools like Wireshark to find the common patterns for an app.

> ***Note:*** *Operator or Field Engineer expertise is needed to analyze pcaps and coming up with patterns.*

1. Open the pcap with Wireshark or any other tool
2. Filter pcaps by HTTP/SSL protocols
3. Look for below fields:
   — Host field in HTTP Get
   — User-Agent in HTTP Get
   — Server Name Indication extension in SSL Client Hello
   — Common Name in SSL Server Certificate
4. Collate the collected data and come up with Common data pattern across pcaps

Script Syntax and Parameter Description

This section provides a sample script based on patterns in .pdl file and also describes parameters used in sample script.

Table 3-1 Script Syntax

Sample Application signature	Sample Protocol signature

`configurations {`
`affirmed-signature-extension : yes;`
`}`
`application "app-https"  group "Social"{`
`signature {`
`payload-search {`
`pattern: "app-https.com";`
`type: HTTPS.client_hello_sni;`
`}`
`result {`
`attribute:"stream" ;`
`processing:continue;`
`}`
`}`
`}`
	`configurations {`
	`affirmed-signature-extension : No;`
	`}`
	`protocol "proto-http"  group "Web"{`
	`signature {`
	`payload-search {`
	`pattern: "proto-http.com";`
	`type: HTTP.host;`
	`}`
	`result {`
	`attribute:" http-connect" ;`
	`processing:stop;`
	`}`
	`}`
	`}`

> ***Note:***
> *- Script file recommended extension is ".pdl".*
> *- Configurations section should be defined only once in the script file.*
> *- More than one signature can be present in the script file.*
> *- Mention only one pattern per signature block, so if there are multiple patterns found for one app or protocol then multiple signatures need to be mentioned in the script file.*
> *- Mention only existing group names in the script file.*

Parameter Description

Table 3-2 Parameter Description

Token	Data Type	Range/Supported Values	Description
affirmed-signature-extension	Boolean	Yes/No	Provides signatures in the script are extensions or improvements to the signatures supported by Affirmed. When having this parameter value as Yes, if matching application/protocol name is not found in the supported list then signature in the script is considered as new application/protocol signature
pattern	string	<>	Pattern found from the pcap analysis
type	enum	HTTP.host HTTP.user_agent HTTPS.client_hello_sni HTTPS.common_name	Indicates supported metadata fields for pattern match
attribute	string	Refer to DPI docs for supported list attributes	One of the supported application/protocol attributes. Only one attribute is supported per signature. Application attributes must be used when adding application signature and Protocol Attribute must be used when adding protocol signature.
processing	enum	stop/continue	Indicates if packets to be passed to for other signatures validation after current signature match

Compiling a Script

This section describes the steps required to compile and validate scripts offline before loading on to MCC.

1. Copy the script compiler and offline signature validation binary (/opt/thoroughbred/dpisig/lib/andsc) from MCM node to Linux machine/VM
2. Copy the script.pdl and corresponding pcap files to the same Linux machine/VM where script compiler is present
3. Compile the script using the below command to validate script syntax and create a bin file with the name mentioned.

```
./andsc -c <script .pdl file path> <output bin file path>
```

> ***Note:*** *Compiler will append "_o" to the new application/protocol names mentioned in the script for differentiation purposes. Example: if the application name is mentioned as "abcd" in the script then it will be updated as "abcd_o".*

Loading a Script on the MCC

This section describes steps required to load scripts on the cluster.

1. Copy validated script binary to /public/images/ directory as “libscript.bin" on active MCM node. Make sure that the file name is “libscript.bin” on the target.
2. Initiate the loading of the copied bin file via below CLI. If there is a failure, validate the bin file and execute the command again.

   ```
   dpiscript install <append | replace>
   ```

   append: To append the new script signatures to existing script signatures on the cluster

   replace: To replace existing script signatures on the cluster with new script signatures
3. Check the signature data loaded on the cluster using the below command. If latest data is not shown up, then validate .pdl file and contact technical support team

   ```
   dpiscript show-dpiscript-loaded
   ```
4. Configure rules, check the statistics for the successfully loaded applications/protocols based on the requirements.
5. To erase all the script signature, execute below command after removing all the protocol-application-rule-group configuration defined for script-based applications/protocols:

   ```
   dpiscript reset
   ```

> ***Note:***
> *- Operator/field engineers have to maintain a database of defined signatures and version them properly to avoid conflicts.*
> *- DPI Script needs to be loaded individually and have to sync on all the User plane clusters.*
> *- Names given for applications and protocols have to be unique.*